

Techniques and Applications of Linear Regression Using Social Media Addiction Data

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Abstract

According to a journal discussing social media addiction, "it has been reported that social media addiction affects approximately 12% of users across social networking sites," [Hou et al. \(2019\)](#). University aged students tend to use their phones and social media more than any other age group. So what would happen to that percentage when you only account for people between the ages of 18 and 24? Starting with this idea, we based our analysis off of a dataset where students from over 100 different countries were surveyed about their own social media experience as well as their personal daily routines and habits. However, we focused on the relationships of those daily routines and habits with their level of addiction to social media.

1 Introduction

1.1 Statement of the problem

Since COVID, social media has become a very important role in every day life. Relationships with social media look different for everyone. For some, it has become a crutch for going between daily tasks, whereas others have made it their career path. When focusing solely on students, now instead of it being a way to take a break from daily stress, it's adding to that stress in more ways than we think. In this study, we wanted to use techniques and applications of linear regression to analyze social media addiction in students and the impact it has on specific factors of their daily lives.

1.2 Relevance of the Problem

According to [Griffiths \(2000\)](#), social media addiction can even be viewed as a form of Internet addiction where individuals can exhibit a compulsion to use social media to excess. This excess can lead to serious social and behavioral issues. With social media addiction becoming more prevalent in different behavioral, addiction, and mental health researches, it felt necessary to analyze its relationships with students as it could help further that research even more.

1.3 Literature review

To help aid in a better understanding of the techniques and applications of linear regression, we relied on several texts. The main text used throughout the study was *An Introduction to Statistical Learning: with Applications in R* by [James et al. \(2013\)](#). This text provided an

introduction to statistical learning, as well as providing a better understanding when applying linear regression to our data. This text was also utilized in many of the fundamentals for our testing as well as running our model through R.

Another text used was *The elements of statistical learning: data mining, inference, and prediction* by [Hastie \(2009\)](#). This text helped with grasping some of the inference and prediction concepts for our study when it came time for testing.

Additional texts that were used included *Extending the linear model with R: generalized linear, mixed effects and nonparametric regression models* by [Faraway \(2016\)](#) and *Applied linear regression models* by [Kutner \(1996\)](#).

To collect more insight on the relevance of our social media data, we utilized *Internet addiction-time to be taken seriously?* by [Griffiths \(2000\)](#) and *Social media addiction: Its impact, mediation, and intervention* by [Hou et al. \(2019\)](#).

1.4 Limitations

Although this data is compelling and interesting to analyze, there are inevitably limitations. Specific to our reported data, given the data is based off self-reported responses, there is always going to be room for potential bias or misreporting. This could impact the results for some of the variables. Another limitation for our data is that it is cross-sectional. Since this data was collected around 2021, the results would vary if taken today or even in 5 years.

An important limitation to note for our study is that the response variable is not continuous and that the normality, when tested, did not work. With having our tests show deviation

from normality, this could affect the reliability of any statistical inference done in our study. The social media addiction scores were measured on a discrete scale (integers between 1 and 10), so it is not continuous. Despite using the traditional regression model for our study, an alternative model may fit the discrete response variable better to explore different components.

The last limitation to be noted is the potential correlation between predictors. This could introduce multicollinearity, which could be something to utilize in a future study.

2 Main Body

2.1 Data Outline

The analysis in this study is based off of a dataset from Kaggle.com with a sample size of 705 observations. The data presents results for different levels of university students from various countries in 2021. A response variable with addiction score to social media (from 1 to 10) is associated with each student. In this study, we wanted to apply a regression model and look at how and what really affected those addiction scores among the data using **RStudio** for applications.

The demographic variables included in the data are age, gender, highest level of academics, and the country from which the students are from. Then the behavioral variables included are their average time spent on social networks each day, their preferred platform, if they feel like it affects their sleep, how many hours on average of sleep they get nightly, mental health status, any conflicts had over social media, and whether or not they are in a relationship. These variables have been used as predictor variables for the regression model.

One notable caveat in this regression model is that the response variable is at a discrete scale, i.e., scores are only integers between 1 to 10. An appropriate model would have been the ordinal logistic regression, however, for this project, we went with applying the traditional regression models to explore different components of regression.

2.2 Exploratory Analysis

We start with exploratory analysis by looking at the scatterplots and getting a general sense of the response-vs-predictor relationships in the data.

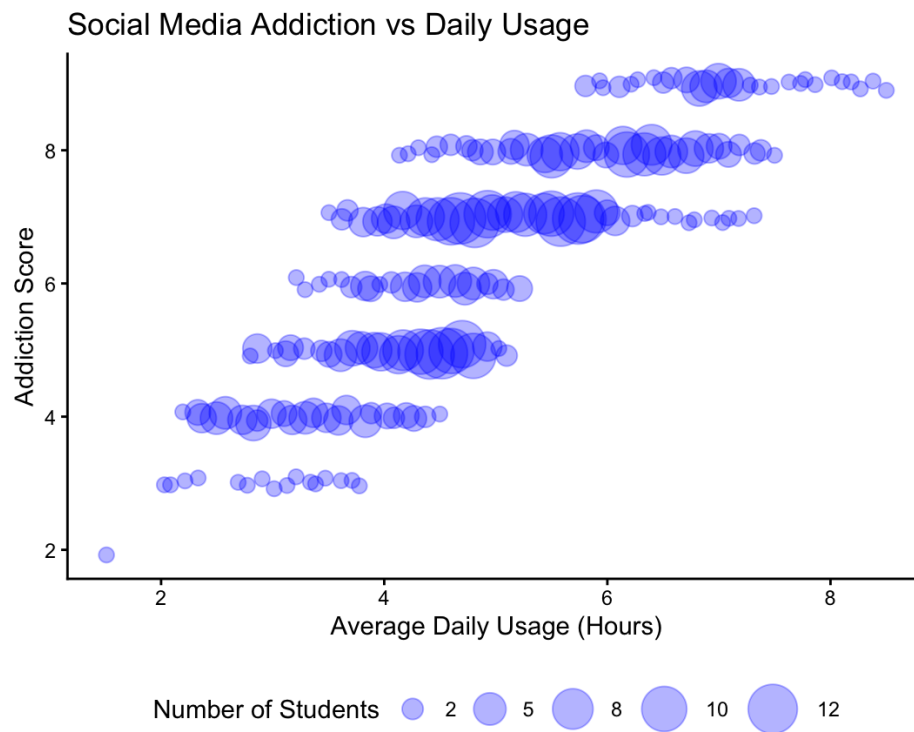


Figure 1: Scatterplot of addiction score and daily social media usage

Figure 1 is a plot between daily social media usage and the response variable. As the logic suggests, the plot shows the same story that the more hours they were spending on social media, they were overall associated with a higher score of addiction. Note that the size of the bubbles are representing the number of students with the same responses. We had to use this due to the discrete nature of the response variable.

Two other scatterplots have been presented in this paper – against the mental health scores (Figure 4) and a reported survey on whether social media affected their academic performance (Figure 5). Figure 4 shows a negative relationship between mental health scores and the

social media addiction score, whereas Figure 5 shows that the group reporting themselves as academically affected were overall associated with higher social media addiction scores.

2.3 Linear Regression

Here, we will formally present the linear regression model, notations, and the assumptions.

2.3.1 Notation and Model

In a multiple linear regression model, we have one dependent variable (denoted with Y) and p many independent variables (denoted with $X_1, \dots, X_p$). Given this setup, the linear regression model establishes a "linear" relationship between the response and the predictor as follows:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \varepsilon_i. \quad (1)$$

Here, $\beta_0, \dots, \beta_p$ are called the regression coefficients. Specifically, β_0 is called the *intercept*, and other coefficients are called the *slope*, corresponding to the variables they are specific to. We can interpret these coefficients as follows:

- (a) The intercept term β_0 represents the mean addiction score (response variable) when all of the predictor variables $X_1, \dots, X_p$ are zero.
- (b) Each slope coefficient β_j is the mean change in the response when the corresponding predictor variable X_j changes by one unit, holding the other variables as constant.

2.3.2 Assumptions

At the end of equation 1, there is another term: ε_i . This is called the *error term* and it captures the individual variation in the response that cannot be explained by the model. Note that statistical models usually have the error terms since we can never predict accurately. For modeling purposes, the error term is associated with the following assumptions:

$$\mathbb{E}[\varepsilon_i] = 0, \tag{2}$$

$$\text{Var}(\varepsilon_i) = \sigma^2, \tag{3}$$

$$\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \text{for } i \neq j. \tag{4}$$

We can interpret the assumptions as the following.

- (a) The first assumption says that the expected value of the error is zero. This is equivalent to saying that there are no systemic over or under estimations of the response variable if we were able to perform the regression model many times.
- (b) The second assumption is about constant variance. The spread, σ^2 , is the same throughout all of the observations. This is also called the *homoscedasticity* assumptions.
- (c) The final term is about independence of the error terms. One subject's error term does not contribute anything to predicting the same of another subject.

In addition, a further assumption is used for inference purposes when the error terms are normally distributed with mean 0 and variance σ^2 , i.e., $\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. This will be important for the hypothesis tests presented in Section 2.4.

2.3.3 Matrix Representation

The solution of a regression problem needs the matrix representation. For the matrix form, we can simultaneously write all the regression equations as the following:

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} 1 & X_{11} & \cdots & X_{1p} \\ 1 & X_{21} & \cdots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{np} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}. \quad (5)$$
$$\iff \mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Here, $\mathbf{Y}$ is the $n \times 1$ vector of responses stacked as a column vector (left hand side of the above equation). The right hand side of the equation collects the predictors into a matrix (often called the design matrix) $\mathbf{X}$. Each row of $\mathbf{X}$ corresponds to one observation, and each column corresponds to one predictor or the intercept. The coefficient vector $\boldsymbol{\beta}$ and the error term $\boldsymbol{\varepsilon}$ are also similarly created as column vectors for the matrix form representation.

2.3.4 Solution

Following the matrix notations, our goal is to get a best estimate of the unknown coefficients $\boldsymbol{\beta}$ and the unknown error variance σ^2 . The idea in regression analysis is to minimize the sum of square of errors, i.e., to minimize the squared differences between the true response and

the model predictions:

$$\begin{aligned}\hat{\beta} &= \arg \min_{\beta} \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi}))^2 \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.\end{aligned}$$

This also helps to express the fitted values of the model as the following:

$$\hat{\mathbf{Y}}_i = \mathbf{X}\hat{\beta} = \hat{\beta}_0 + X_{i1}\hat{\beta}_1 + \dots + X_{ip}\hat{\beta}_p$$

The matrix $\mathbf{X}'\mathbf{X}$ contributes to the variance of $\hat{\beta}$, i.e., we can show that $\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

2.3.5 Our Results

For our data, once the regression model is fitted, we see the regression output presented in Table 3. The first column in the table are the estimated coefficients. As an example, here $\hat{\beta}_0 = 9.79$: the mean predicted response when all the predictors are 0. $\hat{\beta}_{\text{Age}} = -0.06$, i.e., keeping every other predictor the same, if age increases by one year, mean social media addiction is predicted to go down by 0.06 units. We will gradually explain the other columns in the next few tables, as well as the terms like `Most_Used_Platform_Instagram`.

2.4 Hypothesis Testing

In a regression problem, often an important question is to check whether our predictors have meaningful relationship with the response. If there is no relationship, then changes in the predictors do not lead to any changes in the response. This is associated with the case that

the corresponding coefficient β_j is 0. Hence, here we test:

$$H_0 : \beta_j = 0, \quad H_A : \beta_j \neq 0.$$

The test statistic is given by the t statistic:

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}, \quad (6)$$

where $\hat{\beta}_j$ is the estimated coefficient (first column in Table 3) and $\text{SE}(\hat{\beta}_j)$ is its standard error (second column in Table 3). Under H_0 , this statistic follows a t -distribution with degrees of freedom $\text{df} = n - (p + 1)$. Computer programs typically also report the associated p -value – the probability of observing a test statistic at least as extreme as t_j under H_0 . It makes the decision-making easier since smaller p -values indicate an evidence against H_0 , indicating that the predictor helps to explain any variation due to the response. Both the t -statistic and p -value are reported in column 3 and 4 from the regression output table.

As an example, **Age** in Table 3 has fitted coefficient 9.79 and standard error 0.47. Hence, the t -statistic is $\frac{9.79}{0.47} \approx 20.73$. This is already an extremely large number, indicating a p -value approximately zero. Hence, **Age** is a significant predictor in our model.

Among the list of predictors, some of the most significant ones are mental health score ($\hat{\beta} = -0.564$), conflicts over social media ($\hat{\beta} = +0.359$), average daily usage ($\hat{\beta} = +0.164$), and the average amount of sleep ($\hat{\beta} = -0.103$). The impact on academic performance ($\hat{\beta} = +0.751$) also showed to be a significant predictor. However, this variable may not act

independently of the response, so it should be interpreted with regard to potential bias.

Another fundamental hypothesis test here is to check if the overall model is significant or not. This is testing that whether any of the predictor variables is associated with a non-zero coefficient. Here we test:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0,$$

$$H_A : \text{At least one } \beta_j \neq 0,$$

The alternate hypothesis states that at least one of the predictors show a linear associated with the response variable. The test statistic is given by:

$$F = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \frac{n - p - 1}{p}$$

This is the ratio of the explained variation by the model divided by the model degrees of freedom (p), and the error variation divided by the error degrees of freedom $n - p - 1$. For our data, the F-statistic value is 957 with $p = 22$ and $n - p - 1 = 682$ degrees of freedom, a p-value that is approximately zero, and an adjusted R^2 value of 0.9676. This suggests what we may have already guessed from the t -tests; the model is overall significant.

2.5 Normality Tests

Recall that, the error terms ε_i are assumed to be normally distributed. We need to assess this assumption. Here we present two ways to do this.

2.5.1 Q-Q plot

The Q-Q plot compares the theoretical quantiles from a normal distribution against the quantiles from the observed residuals. If our assumptions are correct, both sets of the quantiles should be fairly similar to one another and the points will lie close to a straight reference line. However, as seen in Figure 2, the plot indicates a violation of normality with the huge deviations at the tails. This is likely due to the discrete nature of our response variable.

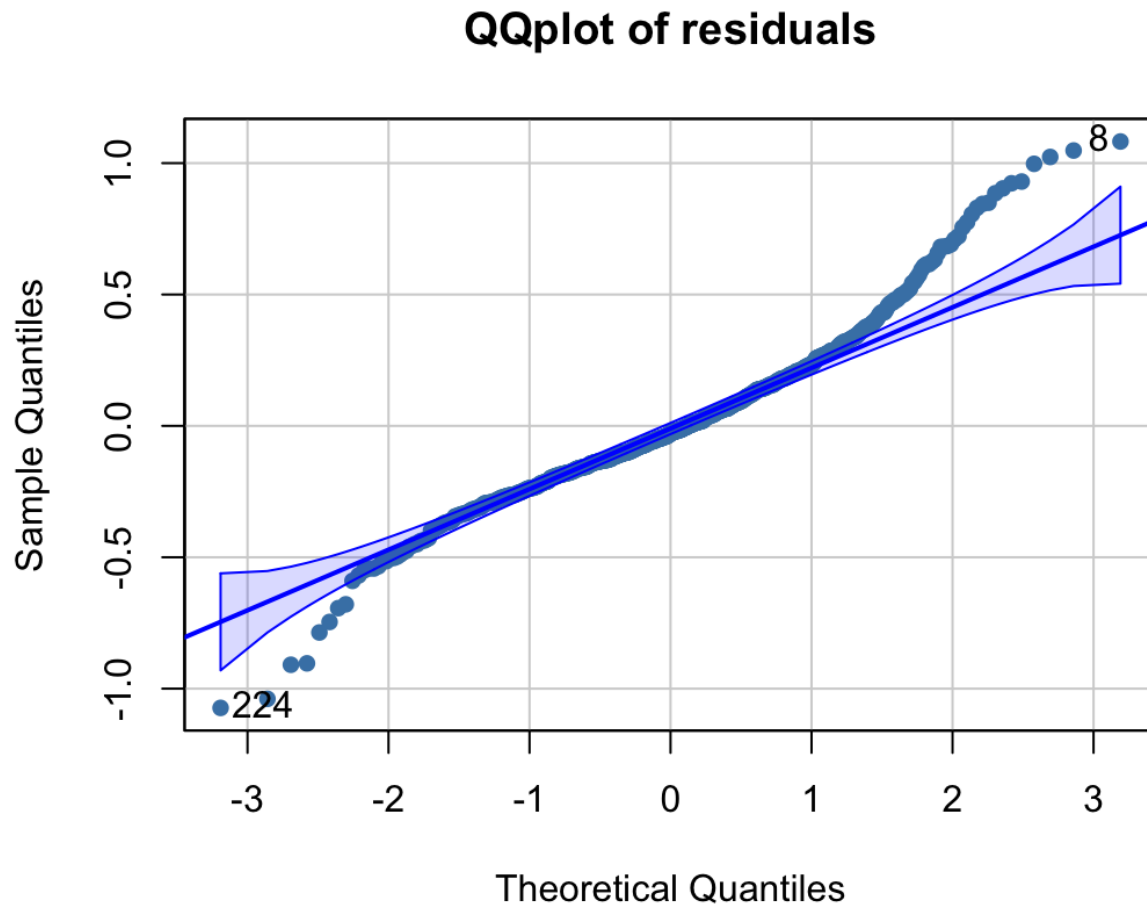


Figure 2: QQplot of the regression residuals

2.5.2 Shapiro-Wilk Test

Instead of relying on a visual check, the Shapiro–Wilk test does a formal checking of the normality assumptions. This test evaluates the null hypothesis of residual normality, and the test statistic measures how closely the ordered sample values align with the expected values under normality.

H_0 :residuals are normally distributed.

H_A :residuals are not normally distributed.

A small p -value in this test indicates evidence against the null hypothesis. This would suggest that the residuals deviate from a normal distribution. A large p -value would indicate that there is insufficient evidence to reject any normality.

Using the Shapiro-Wilk test, the test resulted in a statistic of $W = 0.9625$ with a p -value approximately zero. This provides strong evidence to reject the null hypothesis of normality, which is a problem for our application. However, since the goal of this analysis is to study and apply linear regression methods, we proceed with the model despite this violation.

2.6 Dummy Variables

When the predictors are categorical, there is no meaning of $\beta_j X_j$. In order to include such variables in regression models, we use the dummy variable approach. This is applied to all the predictors including gender, academic level, country, most used platforms, and relationship statues.

We will demonstrate it with one example with the **Relationship Status** variable. In the data, Relationship Status has three variations in responses: **single**, **in a relationship**, and **complicated**. With three categorical levels, we will need $3 - 1 = 2$ dummy variables. One of the three levels are used as the baseline or reference level, typically chosen as the numerical order in R. Here, **Complicated** is chosen as the reference group, and the other two dummy variables are created as:

Relationship Status	D_1 (In Relationship)	D_2 (Single)
Complicated (Baseline)	0	0
In a Relationship	1	0
Single	0	1

Table 1: Dummy variables for relationship status

This way, all the categorical variables are converted into numbers, and hence can be used in the regression models. Note that, the corresponding coefficient of a dummy variable represent the differences in an expected addiction score, $\hat{y}$, relative to it's reference category while holding the other predictors constant. As an example, our model showed that **Relationship_Status_In Relationship** has a coefficient of -0.05 . This is stating that students who reported to be in a relationship have overall 0.05 units lower addiction scores compared to students in the baseline category, i.e., who reported to be in a **Complicated** relationship situation. Similarly, from the output, **Relationship_Status_Single** has a coefficient of -0.16 . This is saying that students who reported to be single, have the lowest addiction score with it being overall 0.16 units lower than the reference group **Complicated**. Table 3 shows the other coefficients

as well corresponding to the dummy variables for `Most_Used_Platform` and `Gender`.

2.7 Train-Test

Our last studied technique is the train-test method. It is well-known that regression, like any statistical model, suffers from *overfitting*. The more data it sees, it tries to memorize the pattern instead of learning. This is why it is essential to artificially create a subset that the model does not use to train (i.e., estimate the unknown coefficients) and then check the predictive performance.

In order to apply this approach, the dataset is split into two subsets: a training set (70% of the data) and a test set (30% of the data). Here, with 705 observations in our data, a 70/30 split would give approximately 490 observations for a training set and 215 for a testing set.

Figure 3 visually shows how the split is applied to the linear regression model.

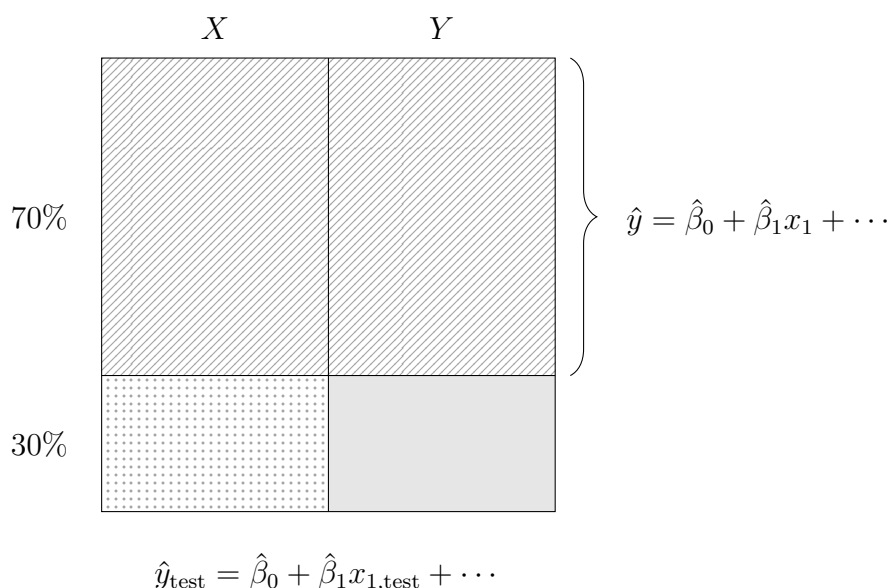


Figure 3: Train-testing splits for regression modeling

The train set is then used to estimate the regression coefficients. After that, we predict the

response for the test set and check how good they are aligning with the true response. For evaluation, we use mean square error (MSE), defined as

$$\text{MSE} = \frac{1}{n_{\text{Test}}} \sum (y_i^{\text{Test}} - \hat{y}_i^{\text{Test}})^2.$$

In our data, the MSE of the test set is 0.0703, with a RMSE (square root of MSE) as 0.27. Our RSME is stating that for our model, on average, the predicted addiction scores will differ from our observed values by roughly 0.27 units. That relatively small prediction error leads us to suggest that the model will generalize reasonably well to unseen data. A visualization of this generalization is visualized in Figure 6. Additionally, a snapshot of the predicted addiction scores and observed values from the test set is listed in Table 2.

	Actual (y)	Predicted ($\hat{y}$)	Residual ($y - \hat{y}$)
1	8.00	7.07	0.93
2	3.00	3.18	-0.18
3	9.00	8.45	0.55
4	4.00	5.04	-1.04
5	7.00	6.68	0.32
6	9.00	9.50	-0.50
7	2.00	2.57	-0.57
8	8.00	6.92	1.08
9	5.00	4.91	0.09
10	4.00	4.75	-0.75

Table 2: Comparison of predicted addiction scores to observed values in the test set.

3 Conclusions

3.1 Summary

Social media addiction affects every important factor in students day to day lives. Through all of our testing and predicting, students appearing to have healthier routines and patterns are resulting in lower addiction scores. While students with higher addiction scores tend to have worse mental health and poorer well-being. Increased time on social media and its interference with students' daily lives is doing more harm than anything else.

3.2 Suggestions for Further Study

Social media addiction and its role in mental health disorders should continue to be studied further. Specifically, assessing any multicollinearity between similar mental health disorders and which are affected by or affect the addictions scores more. Such as depression, anxiety, eating disorders, personality disorders, attention deficit disorders, etc.

In relation to our deviation from normality, different testing to address the normality would be beneficial in a future study to validate the statistical inferences made. With the response variable being not continuous and on a discrete scale, an alternative model for a future study may fit the discrete response variable better to explore different components.

After looking at university students, a comparison of internet and social media addiction in children and people in later adulthood could help further addiction research of all ages. With so many online tools to help learning and growth in toddlers, it would be interesting to see if tablet use could predict addiction scores through their aging.

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Appendix

A Addiction Scores Vs. Mental Health Scores

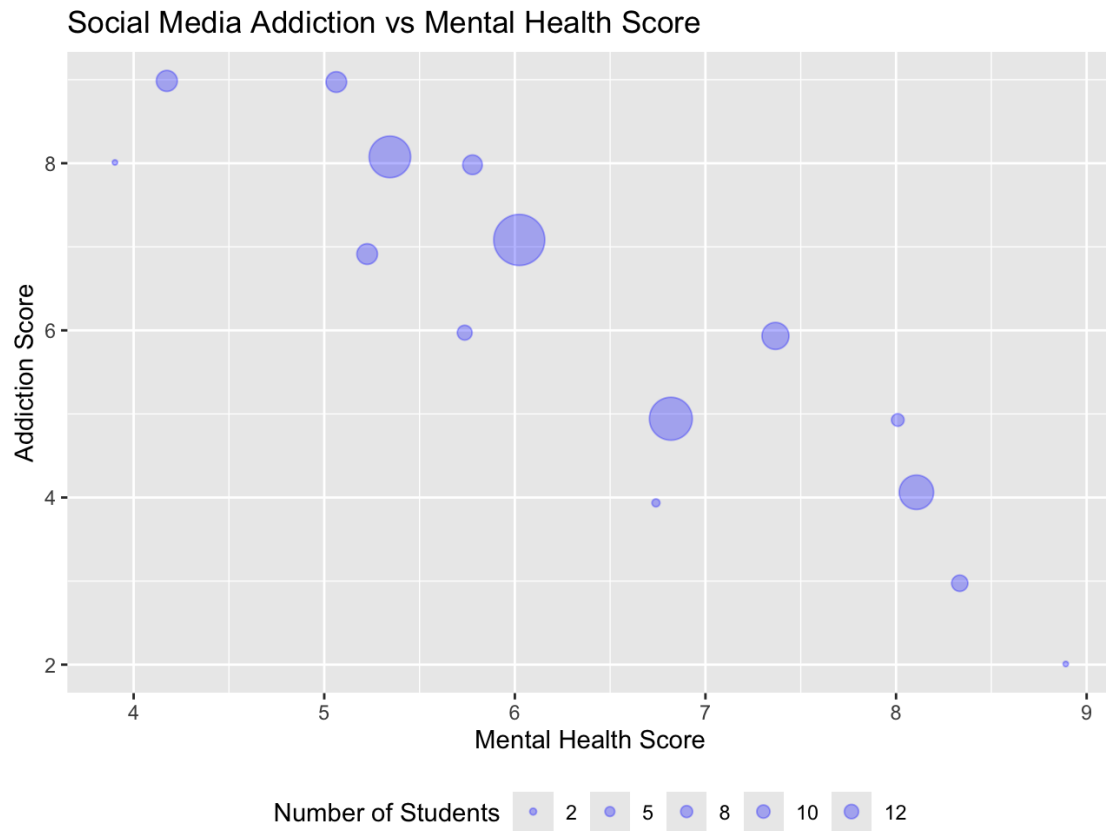


Figure 4: With higher mental-health score (i.e., better mental health conditions), addiction scores tend to decrease.

B Addiction Scores Vs. Affects on Academic Performance

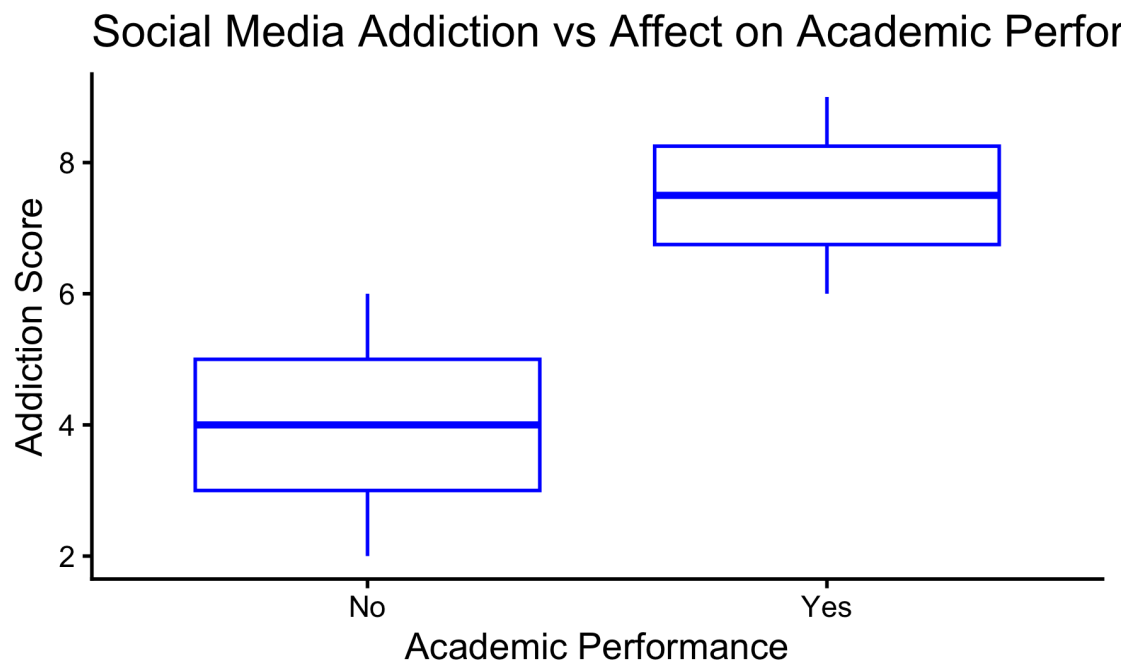


Figure 5: Those who say social media affects their academic performance generally are associated with higher addiction scores.

C Regression model output

Here we present the table obtained from the regression model. It is presented in R's default

style for four values collection – estimate, standard error, t statistic, and the p value.

	term	estimate	std.error	statistic	p.value
1	(Intercept)	9.79	0.47	20.73	0.00
2	Age	-0.06	0.02	-3.62	0.00
3	GenderMale	-0.07	0.03	-2.28	0.02
4	Academic_LevelHigh School	-0.17	0.09	-1.92	0.05
5	Academic_LevelUndergraduate	-0.18	0.05	-3.75	0.00
6	Avg_Daily_Usage_Hours	0.16	0.02	7.37	0.00
7	Most_Used_PlatformInstagram	-0.03	0.04	-0.89	0.38
8	Most_Used_PlatformKakaoTalk	-0.86	0.09	-9.14	0.00
9	Most_Used_PlatformLINE	-0.90	0.09	-9.46	0.00
10	Most_Used_PlatformLinkedIn	-0.04	0.08	-0.47	0.64
11	Most_Used_PlatformSnapchat	0.08	0.09	0.87	0.38
12	Most_Used_PlatformTikTok	0.14	0.04	3.44	0.00
13	Most_Used_PlatformTwitter	-0.20	0.06	-3.37	0.00
14	Most_Used_PlatformVKontakte	-0.04	0.09	-0.41	0.68
15	Most_Used_PlatformWeChat	-0.08	0.08	-0.97	0.33
16	Most_Used_PlatformWhatsApp	-0.26	0.05	-4.82	0.00
17	Most_Used_PlatformYouTube	0.48	0.10	4.71	0.00
18	Affects_Academic_PerformanceYes	0.75	0.05	14.79	0.00
19	Sleep_Hours_Per_Night	-0.10	0.02	-5.16	0.00
20	Mental_Health_Score	-0.56	0.02	-22.59	0.00
21	Relationship_StatusIn Relationship	-0.05	0.06	-0.90	0.37
22	Relationship_StatusSingle	-0.16	0.06	-2.64	0.01
23	Conflicts_Over_Social_Media	0.36	0.03	11.25	0.00

Table 3: Regression results for social media addiction modeling

D Test Performance

Here we present the plot of the predicted response on the test data. Overall, the positive association shows that the model generalized well on the test data. Note that due to the discrete nature of the response, we decided to use boxplots of the predicted values at each discrete level of the true response.

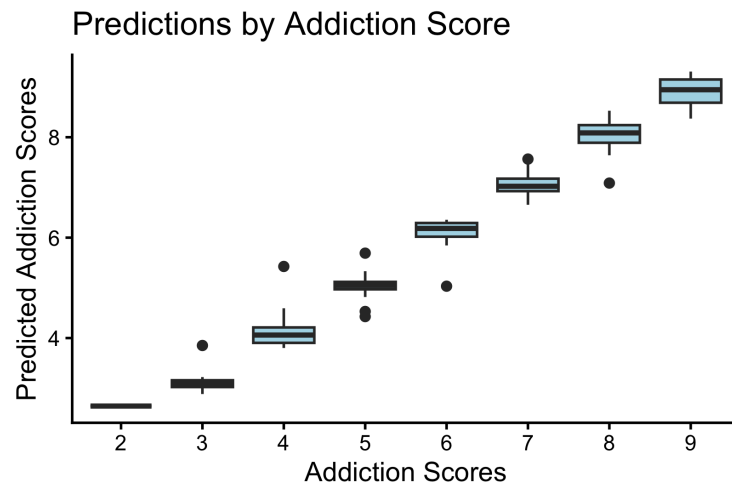


Figure 6: Test data prediction results